



Efficacy and safety of retatrutide, a GIP, GLP-1, and glucagon receptor agonist, in people with type 2 diabetes and inadequate glycaemic control with diet and exercise (TRANSCEND-T2D-1): a double-blind, randomised, phase 3 trial

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Summary

Background Retatrutide is a GIP, GLP-1, and glucagon triple hormone receptor agonist, under clinical development for type 2 diabetes, obesity, and related complications. We aimed to assess the efficacy and safety of retatrutide as a monotherapy in people with type 2 diabetes that is inadequately controlled by diet and exercise alone.

Methods In this 40-week, phase 3, randomised, double-blind, placebo-controlled trial at 48 sites in the USA, Mexico, and India, we recruited adults (aged ≥ 18 years) with type 2 diabetes that is inadequately controlled by diet and exercise alone, glycated haemoglobin (HbA_{1c}) between 7.0% and 9.5% (53–80 mmol/mol), and BMI of at least 23 kg/m². Participants were randomly assigned (1:1:1:1) to receive retatrutide (4 mg, 9 mg, or 12 mg) or placebo by once-weekly subcutaneous injection. The primary endpoint was the change in HbA_{1c} concentration from baseline to week 40. A key secondary endpoint was the percentage change in bodyweight from baseline to week 40. This trial is registered with ClinicalTrials.gov, NCT06354660, and is completed.

Findings Between April 10, 2024, and April 21, 2025, 930 participants were screened and 537 (296 [55%] female and 241 [45%] male) were randomly assigned: 134 to retatrutide 4 mg, 133 to retatrutide 9 mg, 136 to retatrutide 12 mg, and 134 to placebo. Baseline mean age was 48.8 years (SD 12.1), mean HbA_{1c} concentration was 7.9% (SD 1.1), mean duration of diabetes was 2.5 years (SD 4.4), and mean BMI was 35.8 kg/m² (SD 7.0). 490 (91%) participants completed the treatment period on study drug and 504 (94%) completed the study. For the treatment regimen estimand, the mean change from baseline in HbA_{1c} concentration was –1.69% (SE 0.11) with retatrutide 4 mg, –1.86% (0.10) with 9 mg, and –1.94% (0.08) with 12 mg, versus –0.81% (0.12) with placebo, resulting in estimated treatment differences versus placebo of –0.88% (95% CI –1.18 to –0.59) with retatrutide 4 mg, –1.04% (–1.32 to –0.76) with 9 mg, and –1.12% (–1.39 to –0.85) with 12 mg (all $p < 0.0001$). The mean percentage change from baseline in bodyweight was –11.5% (SE 0.7) with retatrutide 4 mg, –13.9% (0.8) with 9 mg, and –15.3% (0.8) with 12 mg, versus –2.6% (0.5) with placebo. The most frequent adverse events with retatrutide were generally mild to moderate gastrointestinal events, which subsided over time. Study intervention discontinuations due to adverse events were 2–5% with retatrutide and 0% with placebo. No severe hypoglycaemia was reported. Two deaths occurred during the study, both in the retatrutide 4 mg group and unrelated to the study drug.

Interpretation Retatrutide showed significant improvements in glycaemic control and bodyweight reduction as a monotherapy in adults with type 2 diabetes that is inadequately controlled with diet and exercise alone, with an adverse event profile consistent with molecules with GLP-1 agonist activity, supporting its potential as an effective treatment for type 2 diabetes.

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Introduction

Type 2 diabetes is characterised by hyperglycaemia caused by insulin resistance and a decline in pancreatic β -cell function.¹ Chronic progression of type 2 diabetes leads to macrovascular and microvascular complications, including cardiovascular disease.¹ The American Diabetes Association recommends a glycated

haemoglobin (HbA_{1c}) target of less than 7% (<53 mmol/mol) for most individuals but endorses more stringent HbA_{1c} targets (eg, <6.5% [<48 mmol/mol]) for individuals with shorter disease duration, good overall health, longer life expectancy, and low treatment-related risks, particularly hypoglycaemia.² Excess adiposity, especially visceral fat and ectopic fat deposition in liver

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Research in context

Evidence before this study

The American Diabetes Association standards of care recommend a target glycated haemoglobin (HbA_{1c}) of less than 6.5% for individuals with type 2 diabetes who are in good health and have low treatment risks. Furthermore, the standards of care suggest that a sustained bodyweight loss of greater than 10% could have disease-modifying effects and possibly lead to remission of diabetes. We searched PubMed on March 13, 2026, using the terms “glucagon-like peptide 1 receptor agonist”, “glucose-dependent insulintropic polypeptide”, “glucagon”, “retatrutide”, “triple agonist”, “type 2 diabetes”, “obesity”, and “chronic weight management”, with no date or study duration restrictions. The results of the search showed that GLP-1 monoreceptor agonists and a dual GIP and GLP-1 receptor agonist have shown robust weight reduction efficacy in individuals with type 2 diabetes. The GIP and GLP-1 receptor agonist tirzepatide has shown superior effects on glycaemic control and bodyweight compared with selective GLP-1 receptor agonism (dulaglutide and semaglutide). However, the weight reduction efficacy of these agents is generally attenuated in trial participants with type 2 diabetes compared with participants without type 2 diabetes, highlighting an unmet need for therapies delivering robust weight reduction alongside glycaemic control. Additionally, treatment of hyperglycaemia without suitable weight reduction can leave people with type 2 diabetes at a high risk of cardiometabolic complications associated with obesity and overweight, such as hypertension, dyslipidaemia, metabolic dysfunction-associated steatotic liver disease, and cardiovascular disease. Retatrutide, a GIP, GLP-1, and glucagon triple hormone agonist, might further enhance weight reduction, as targeting the glucagon receptor has been shown to increase energy expenditure, suppress appetite, and enhance fat metabolism while providing meaningful improvement in glycaemic control. In a phase 2 clinical trial in 281 people with type 2 diabetes (NCT04867785), treatment with retatrutide resulted in greater bodyweight reduction compared with a selective GLP-1 receptor agonist (dulaglutide).

Added value of this study

TRANSCEND-T2D-1 is the first phase 3 clinical trial of retatrutide for the treatment of type 2 diabetes. We investigated the

efficacy and safety of retatrutide (4 mg, 9 mg, or 12 mg) compared with placebo as a monotherapy for 40 weeks. Treatment with retatrutide resulted in significant reductions in HbA_{1c} and fasting serum glucose compared with placebo. The proportion of participants reaching glycaemic targets of less than 7.0%, less than or equal to 6.5%, and less than 5.7% were also significant for all tested retatrutide doses at week 40. Treatment with retatrutide also resulted in significant bodyweight reduction, beyond that observed in similar studies of molecules with GLP-1 agonist activity. The composite endpoints of reaching the target of HbA_{1c} concentration of 6.5% or less and the potentially disease-modifying threshold of weight reduction of 10% or more was reached by more than half of participants receiving retatrutide, showing the efficacy of retatrutide in improving glycaemic control while simultaneously driving clinically meaningful weight reduction. Retatrutide also improved other cardiometabolic outcomes, including lipid profile and blood pressure. The safety profile was overall consistent with molecules with GLP-1 agonist activity, with the most commonly reported adverse events being gastrointestinal in nature. Together, TRANSCEND-T2D-1 showed that triple hormone receptor agonist retatrutide could deliver robust weight reduction alongside glycaemic control in early type 2 diabetes, potentially addressing an unmet need for effective therapies that act on both endpoints simultaneously for people with type 2 diabetes.

Implications of all the available evidence

The results of the TRANSCEND-T2D-1 study show potent glucose-lowering effects of retatrutide in people with early type 2 diabetes, accompanied by significant bodyweight reduction. The safety results were similar to those reported for molecules with GLP-1 agonist activity. This first phase 3 trial of retatrutide in type 2 diabetes supports the use of retatrutide as a monotherapy for early type 2 diabetes. Retatrutide is being investigated in the additional TRANSCEND-T2D clinical trials for use in different populations with type 2 diabetes, including in direct comparison with other available therapies.

and skeletal muscle, can lead to insulin resistance and is often associated with type 2 diabetes.³⁻⁶ Therefore, weight management is important for controlling and treating type 2 diabetes.⁷ Although 5-7% weight loss improves blood glucose levels, reaching and maintaining a greater than 10-15% weight loss is associated with greater benefits, including improved cardiovascular outcomes and possible remission of type 2 diabetes.^{7,8}

GLP-1 receptor agonist-based drugs are an effective treatment option for people with type 2 diabetes, associated with improvements in glycaemic control and notable weight reduction.⁹⁻¹¹ GLP-1 receptor

agonist-based drugs have shown efficacy across a vast range of populations with type 2 diabetes, including as a monotherapy in people with type 2 diabetes treated only with diet and exercise, where reductions in HbA_{1c} up to 2.07% and bodyweight reductions up to 11.0% were observed after 40 weeks with the dual GIP and GLP-1 receptor agonist tirzepatide.^{12,13} Robust weight reduction has been observed in people with type 2 diabetes treated with GLP-1 monoreceptor agonists or a dual GIP and GLP-1 receptor agonist. However, the extent of weight reduction conferred by these agents is generally attenuated in people with type 2 diabetes compared with

people without type 2 diabetes. 10% or more weight reduction was seen in 66% of participants with type 2 diabetes versus 86% without in the STEP-UP (semaglutide 7·2 mg) study, and in 70% of participants with type 2 diabetes versus 90% without in the SURMOUNT (tirzepatide 15 mg) study (efficacy estimand).^{14–17} These results highlight an unmet need for therapies delivering robust weight reduction alongside glycaemic control for people with type 2 diabetes. Addition of glucagon receptor targeting could increase bodyweight reduction by suppressing appetite, modulating substrate use in the liver, and increasing energy expenditure, leading to greater effects than mono-incretin or dual-incretin receptor agonists.^{18,19}

Retatrutide (LY3437943) is a novel GIP, GLP-1, and glucagon receptor agonist, which is administered via once-weekly subcutaneous injection.^{20,21} In a phase 2, double-blind, double-dummy, 36-week study among 281 US participants with type 2 diabetes treated with diet and exercise alone (28% of participants) or with a stable dose of metformin (72% of participants), retatrutide at doses of 8 mg or 12 mg once weekly resulted in greater reductions in HbA_{1c} (up to 2·16%) and bodyweight (up to 16·9%) compared with the GLP-1 receptor agonist dulaglutide at a 1·5 mg dose.²¹

In the TRANSCEND-T2D-1 clinical trial, we aimed to assess the efficacy and safety of once-weekly retatrutide as a monotherapy in people with type 2 diabetes that is inadequately controlled with diet and exercise alone.

Methods

Study design

This 40-week, phase 3, multicentre, randomised, parallel-arm, double-blind, placebo-controlled study was held at 48 medical research centres and hospitals in India, Mexico, and the USA. Written informed consent was obtained from each participant before they entered the study. The protocol was approved by local institutional review boards at each participating centre (appendix pp 3–4), and the clinical trial was done in accordance with the International Council for Harmonisation E6 guidelines for Good Clinical Practice and with the Declaration of Helsinki guidelines on good clinical practices. This trial is registered with ClinicalTrials.gov, NCT06354660, and is completed.

Participants

Eligible participants were adults (aged ≥18 years) with type 2 diabetes, HbA_{1c} of between 7·0% (53 mmol/mol) and 9·5% (80 mmol/mol) despite diet and exercise treatment, and BMI of 23·0 kg/m² or higher. Participants had not used any oral or injectable antihyperglycaemic medications during the 90 days preceding screening or during screening, were naive to insulin therapy except for gestational diabetes or up to 14 days of use for acute treatment, and had stable weight (self-reported change ≤5 kg) for at least 90 days before screening. Key exclusion

criteria were type 1 diabetes, severe hypoglycaemia or hypoglycaemia unawareness within the last 6 months, diabetic retinopathy or macular oedema requiring treatment, history of pancreatitis, estimated glomerular filtration rate of less than 15 mL/min per 1·73 m², or severe psychiatric disorder. A complete list of eligibility criteria is available in the appendix (pp 5–9). Sex (male or female), race, and ethnicity were self-reported.

Randomisation and masking

Participants were randomly assigned (1:1:1) to receive 4 mg, 9 mg, or 12 mg of retatrutide, or placebo, by a computer-generated random sequence using an interactive web-response system, stratified by country, HbA_{1c} (≤8·0% or >8·0% [≤ 64 mmol/mol or >64 mmol/mol] at screening), and previous use of any antihyperglycaemic medication (yes or no). Investigators, site staff, clinical monitors, study team, and participants remained masked to treatment assignments until study completion.

Procedures

After a screening period of up to 4 weeks, participants self-administered a once-weekly subcutaneous injection of retatrutide or placebo with a multidose prefilled pen (KwikPen; Eli Lilly and Company, Indianapolis, IN, USA), for a treatment period of 40 weeks, followed by a treatment-free safety follow-up period of 4 weeks (appendix p 38). Participants assigned to retatrutide followed a dose escalation scheme, starting at 2 mg of retatrutide and increasing the dose every 4 weeks, until the randomly assigned maintenance dose of 4 mg, 9 mg, or 12 mg was reached. Participants assigned to 4 mg and 9 mg of retatrutide, or to placebo, followed a sham dose-escalation schedule matching the 12 mg retatrutide group to maintain masking. Participants were counselled to continue their usual exercise habits and follow a healthy meal plan (with an emphasis on consuming healthy macronutrients and staying hydrated) throughout the study; however, diet and exercise were not monitored in this trial.

Dose modification was only allowed for the management of intolerable gastrointestinal symptoms, poor nutrition or dehydration, the participant reaching a BMI of 20 kg/m² or less, or perceived excessive weight reduction (appendix pp 11–12). Management of intolerable gastrointestinal symptoms is described in the appendix (pp 11–12).

Rescue therapy with antihyperglycaemic medication was allowed for severe persistent hyperglycaemia according to specific criteria (appendix p 10). Use of GLP-1 receptor agonists or other incretin-based therapies, amylin analogues, or DPP-4 inhibitors was not permitted.

Ophthalmological evaluation of fundus photographs for diabetic retinopathy or macular oedema was carried out for all participants at baseline and week 40, with an additional assessment at week 24 for those found to have

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any degree of diabetic retinopathy or macular oedema at baseline (appendix pp 12–13).

No dedicated data monitoring committee was used because, at the phase 3 programme level, an independent data monitoring committee regularly reviewed unmasked safety data from the TRIUMPH-2 (NCT05929079) and TRIUMPH-3 (NCT05882045) type 2 diabetes studies, with any relevant recommendations implemented across the TRANSCEND-T2D programme. TRANSCEND-T2D-1 was not within the scope of routine scheduled data monitoring committee reviews and was monitored through masked trial-level safety reviews, with no circumstances arising during the study that warranted escalation to unmasked data monitoring committee review per the data monitoring committee charter.

Outcomes

The primary endpoint was change in HbA_{1c} from baseline to 40 weeks, for retatrutide 4 mg, 9 mg, and 12 mg compared with placebo. Key secondary endpoints controlled for type I error at week 40 for retatrutide 4 mg, 9 mg, and 12 mg versus placebo were: incidence of HbA_{1c} less than 7·0% (53 mmol/mol) and less than or equal to 6·5% (48 mmol/mol); change and percentage change in bodyweight; incidence of bodyweight reduction of 5% or more and 10% or more from baseline; the composite incidence of HbA_{1c} of 6·5% (48 mmol/mol) or lower and bodyweight reduction of 10% or more; and—for retatrutide 9 mg and 12 mg versus placebo—incidence of HbA_{1c} of less than 5·7% (39 mmol/mol), change in fasting serum glucose, incidence of bodyweight reduction of 15% or more, percentage change in non-HDL cholesterol and triglycerides, and change in systolic blood pressure. Additional secondary endpoints not controlled for type I error at week 40 for retatrutide 4 mg, 9 mg, and 12 mg versus placebo included: change in daily average 7-point self-monitored blood glucose (SMBG); waist circumference; BMI; diastolic blood pressure; incidence of bodyweight reduction of 20% or more and 25% or more; percentage change in total, HDL, LDL, and VLDL cholesterol, apolipoprotein B, and high-sensitivity C-reactive protein; and, for retatrutide 4 mg, incidence of HbA_{1c} of less than 5·7% (39 mmol/mol), change in fasting serum glucose, incidence of bodyweight reduction of 15% or more, percentage change in non-HDL cholesterol and triglycerides, and change in systolic blood pressure. All secondary endpoints are listed in the appendix (pp 20–21). Safety was assessed throughout the trial, including serious adverse events, treatment-emergent adverse events, and discontinuations due to adverse events (appendix pp 20–21).

Statistical analysis

The statistical analysis methods were prespecified in the statistical analysis plan (appendix pp 15–19). We estimated that 480 participants would provide at least 90% power to assess the superiority of retatrutide 4 mg,

9 mg, and 12 mg, each tested in parallel against placebo at the two-sided significance level of 0·0167 (0·05 overall α split between comparisons) with a two-sample *t* test based on the primary endpoint (change from baseline in HbA_{1c} at week 40), with a treatment effect of –0·6% and a common SD of 1·2%. These assumptions ensured power under both the treatment regimen and efficacy estimands.

All randomly allocated participants were included in the efficacy analyses. Two estimands were prespecified: the treatment-regimen estimand and the efficacy estimand. The treatment-regimen estimand assessed the estimated average treatment effect regardless of adherence to the study intervention or initiation of additional antihyperglycaemic medications. For the treatment regimen estimand, continuous variables were imputed for missing values and then analysed using an ANCOVA model with a variance estimator robust to model misspecification and heteroscedasticity.^{22,23} Binary variables were assessed using a logistic regression model. The models included treatment group, country, and a single strata variable for other stratification variables (baseline HbA_{1c} [$\leq 8\cdot0\%$ or $>8\cdot0\%$] and previous use of any antihyperglycaemic medication [yes or no]) as factor variables, baseline value, interaction between treatment group and strata variable, and interaction between treatment group and baseline value. For subgroup analyses under the treatment regimen estimand, the ANCOVA model was fitted separately within each category of subgroup. The model-based estimated means and variance–covariance estimates from these separate models were used to test the treatment-by-subgroup interaction at a significance level of 0·10.

The efficacy estimand assessed the estimated average treatment effect under the assumption that all participants adhered to the assigned study intervention for 40 weeks without using additional antihyperglycaemic medications (>14 days). For the efficacy estimand, continuous variables were assessed using a restricted maximum likelihood-based mixed-model repeated-measures (MMRM) analysis with dose pooling.²⁴ The MMRM included treatment group, visit, country, and a single strata variable for other stratification variables (baseline HbA_{1c} [$\leq 8\cdot0\%$ or $>8\cdot0\%$] and previous use of any antihyperglycaemic medication [yes or no]) as factor variables, baseline value, interaction between visit and treatment group, interaction between visit and strata variable, and interaction between visit and baseline value. If the data did not warrant the use of an MMRM model, such as for endpoints assessed at a single post-baseline timepoint only, then an ANCOVA model was used, with missing values imputed using multiple imputation under the missing-at-random assumption. Binary variables were assessed using the same logistic regression model as that for the treatment regimen estimand, but using multiple imputation under the missing-at-random assumption.

A graphical testing procedure was applied under each estimand to strongly control the overall family-wise error rate at a two-sided α level of 0.05 for the primary and key secondary endpoints (appendix p 39). CIs were not adjusted for multiplicity and should not be used for hypothesis testing.

Unless otherwise specified, safety and tolerability assessments were guided by an estimand comparing the safety of retatrutide treatment groups with placebo for the entire study period (the treatment period plus safety follow-up), irrespective of intercurrent events. Hypoglycaemia events occurring after the start of additional antihyperglycaemic therapy were excluded. For categorical safety measures, analyses were conducted using Fisher's exact test or Pearson's χ^2 test. For continuous safety measures, analyses were done using an MMRM, with treatment group and visit as factor variables, baseline value, and the interaction between visit and treatment group. If an MMRM was not warranted, an ANCOVA with treatment group as a fixed effect and the continuous baseline value as a covariate was used. No explicit imputation was done for safety measures.

Role of the funding source

The study sponsor (Eli Lilly and Company) was involved in the study design, data collection, data review, analyses and drafting of the report.

Results

Between April 10, 2024, and Feb 20, 2026, 930 participants were assessed for eligibility and 537 were randomly assigned: 134 to retatrutide 4 mg, 133 to retatrutide 9 mg, 136 to retatrutide 12 mg, and 134 to placebo. Of those randomly allocated, 37 (9%) participants prematurely discontinued retatrutide and ten (7%) participants discontinued placebo. The most common reason for premature study intervention discontinuation was an adverse event in the retatrutide 9 mg and 12 mg groups, and withdrawal by the participant in the retatrutide 4 mg and placebo groups. 94–99% of participants reached their randomly assigned dose of retatrutide following protocol-defined dose escalation (appendix p 22). In this trial, two participants underwent temporary dose de-escalation due to intolerable gastrointestinal symptoms, and two underwent permanent dose reductions. Overall, trial

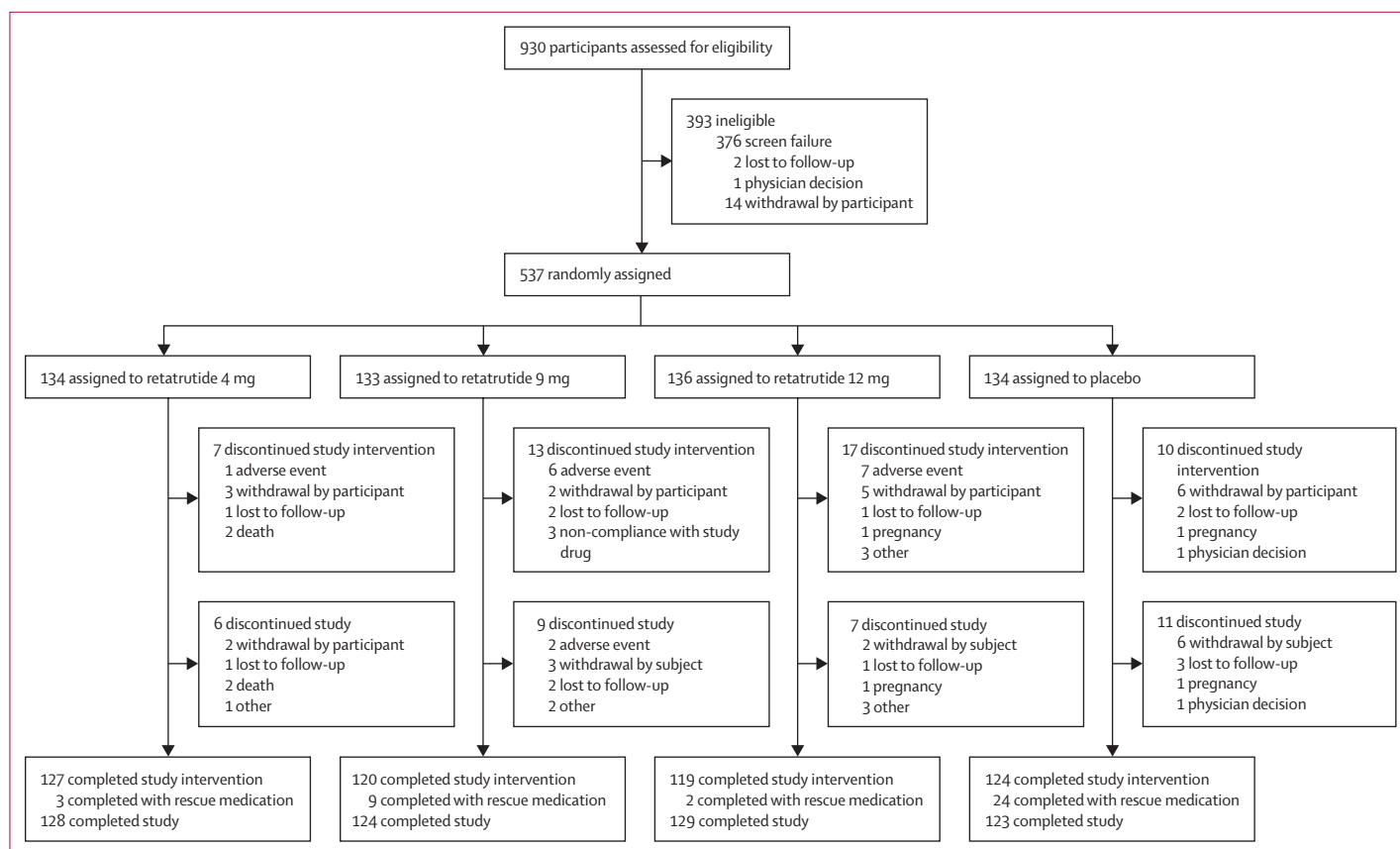


Figure 1: Trial profile

Participant disposition in the TRANSCEND-T2D-1 study. Study completion is defined as completing both the 40-week treatment period and the safety follow-up period. If participants discontinued the study during the treatment period, they also discontinued the study intervention, either earlier or at the same time.

participants were able to tolerate retatrutide without a need for de-escalation or permanent dose reduction. In total, 504 (94%) participants completed the study, with 490 (91%) participants completing 40 weeks on treatment (figure 1).

Demographics and clinical characteristics were similar across treatment groups. At baseline, study participants had a mean age of 48·8 years (SD 12·1), mean HbA_{1c} concentration of 7·9% (SD 1·1; 63·2 mmol/mol [SD 11·5]), mean bodyweight of 96·9 kg (SD 22·5), mean BMI of 35·8 kg/m² (SD 7·0), and mean duration of type 2 diabetes of 2·5 years (SD 4·4). 296 (55%) participants were female and 241 (45%) were male, and 455 (85%) participants had no previous use of antihyperglycaemic medication (table 1).

TRANSCEND-T2D-1 met all prespecified and multiplicity-controlled efficacy endpoints, under both estimands (treatment regimen and efficacy). Data presented are based on the treatment regimen estimand unless otherwise indicated. Results were consistent for the efficacy estimand (appendix pp 25–26).

At week 40, from an overall mean baseline HbA_{1c} of 7·9%, mean change in HbA_{1c} was –1·69% (SE 0·11) with retatrutide 4 mg, –1·86% (0·10) with 9 mg, and –1·94% (0·08) with 12 mg, versus –0·81% (0·12) with placebo (figure 2A, appendix pp 23–24). Retatrutide doses showed significantly greater reduction in HbA_{1c} from baseline to week 40 versus placebo (figure 2A, B). Estimated treatment differences versus placebo were –0·88% (95% CI –1·18 to –0·59) for retatrutide 4 mg, –1·04% (–1·32 to –0·76) for 9 mg, and –1·12% (–1·39 to –0·85) for 12 mg, with $p < 0·0001$ for all comparisons (figure 2A, appendix pp 23–24). HbA_{1c} targets of less than 7·0%, less than or equal to 6·5%, and less than 5·7% were reached by a higher proportion of participants treated with retatrutide at week 40 versus placebo (figure 2E). Similar results were observed for the efficacy estimand (appendix pp 25–26, 40). Exploratory subgroup analyses of the change in HbA_{1c} are shown in the appendix (pp 41–42).

At week 40, the mean change from baseline in fasting serum glucose was –30·1 mg/dL (SE 4·2) with retatrutide 4 mg, –31·9 mg/dL (3·8) with 9 mg, and –30·8 mg/dL (3·6) with 12 mg, versus –1·0 mg/dL (4·5) with placebo (figure 2C, D). Changes in the 7-point SMBG profiles with retatrutide and placebo are shown in figure 2F (efficacy estimand).

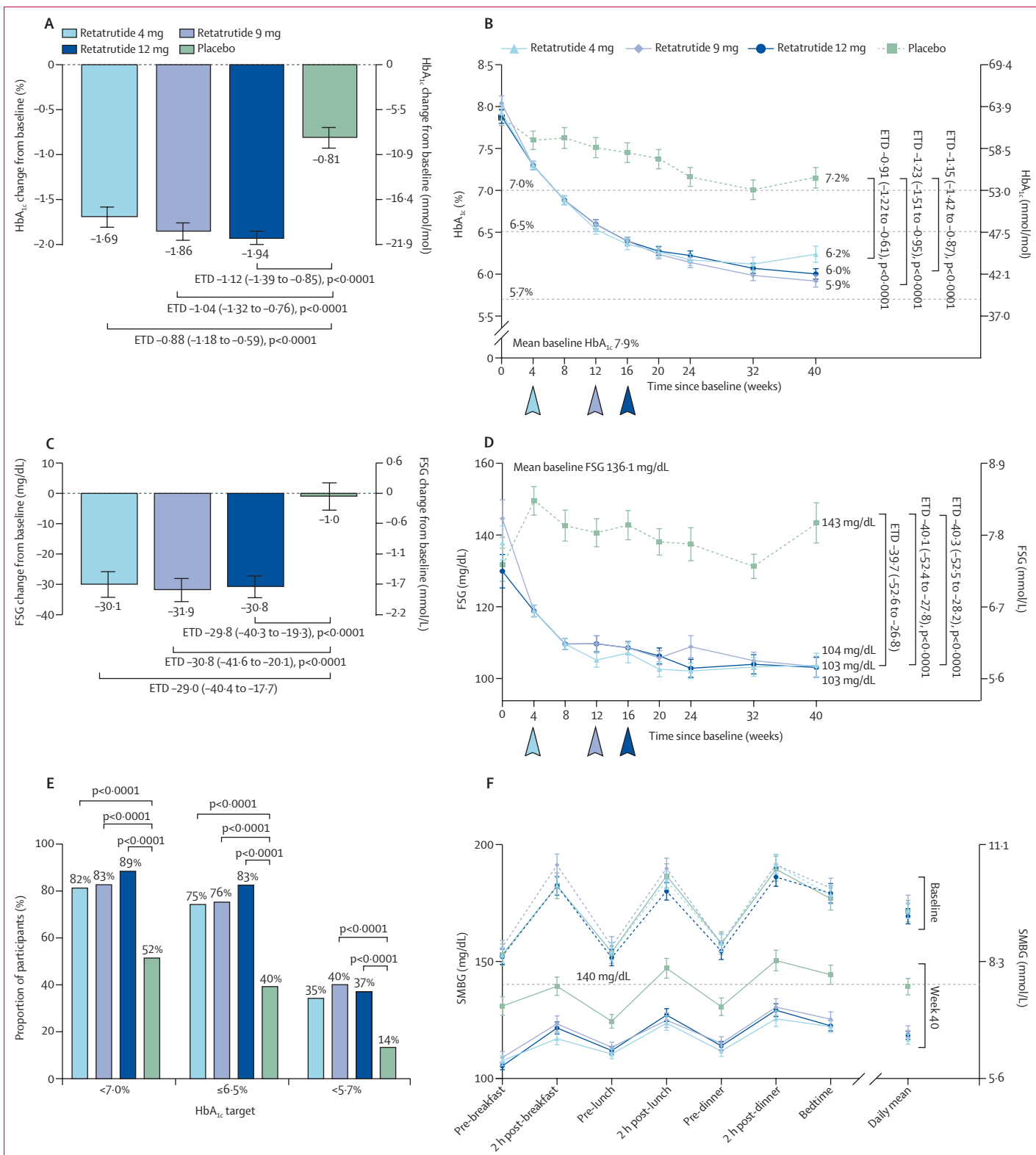
Bodyweight decreased from a mean baseline of 96·9 kg, with mean percentage changes of –11·5% (SE 0·7; –11·1 kg [SE 0·7]) with retatrutide 4 mg, –13·9% (0·8; –13·5 kg [0·8]) with 9 mg, and –15·3% (0·8; –15·1 kg [0·8]) with 12 mg, compared with –2·6% (0·5; –2·7 kg [0·5]) with placebo (figure 3A, B, appendix p 43). Estimated treatment differences versus placebo were –8·9% (95% CI –10·5 to –7·2) for retatrutide 4 mg, –11·3% (–13·1 to –9·5) for 9 mg, and –12·7% (–14·4 to –11·0) for 12 mg (all $p < 0·0001$; figure 3A). Bodyweight reduction had not plateaued by the end of

the 40-week treatment period. A greater proportion of participants treated with retatrutide versus placebo had bodyweight reductions of 5% or more (75–85% vs 26%), 10% or more (52–70% vs 7%), and 15% or more (31–51% vs 3%; figure 3C). Furthermore, a greater proportion of participants treated with retatrutide reached the composite endpoint of HbA_{1c} less than or equal to 6·5% and bodyweight reduction of 10% or more at week 40

	Retatrutide 4 mg (n=134)	Retatrutide 9 mg (n=133)	Retatrutide 12 mg (n=136)	Placebo (n=134)
Age, years	47·9 (12·2)	50·2 (12·1)	49·2 (12·1)	48·0 (12·2)
Sex				
Female	71 (53%)	79 (59%)	80 (59%)	66 (49%)
Male	63 (47%)	54 (41%)	56 (41%)	68 (51%)
Race				
American Indian or Alaska Native	50 (37%)	50 (38%)	49 (36%)	45 (34%)
Asian	41 (31%)	41 (31%)	43 (32%)	45 (34%)
Black or African American	5 (4%)	5 (4%)	8 (6%)	9 (7%)
White	36 (27%)	33 (25%)	35 (26%)	35 (26%)
Ethnicity				
Hispanic or Latino	72 (54%)	71 (53%)	71 (52%)	69 (51%)
Not Hispanic or Latino	62 (46%)	62 (47%)	65 (48%)	65 (49%)
Country				
India	41 (31%)	40 (30%)	41 (30%)	42 (31%)
Mexico	54 (40%)	54 (41%)	55 (40%)	53 (40%)
USA	39 (29%)	39 (29%)	40 (29%)	39 (29%)
HbA _{1c} concentration				
%	7·9 (1·1)	8·0 (1·0)	7·9 (1·0)	7·9 (1·1)
mmol/mol	63·1 (11·9)	64·4 (11·3)	62·7 (10·8)	62·5 (12·1)
HbA _{1c} category				
≤8·0%	79 (59%)	70 (53%)	90 (66%)	85 (63%)
>8·0%	55 (41%)	63 (47%)	46 (34%)	49 (37%)
Fasting serum glucose concentration				
mg/dL	137·8 (52·6)	144·7 (59·6)	129·9 (53·3)	131·8 (51·8)
mmol/L	7·7 (2·9)	8·0 (3·3)	7·2 (3·0)	7·3 (2·9)
Duration of type 2 diabetes, years	2·3 (4·3)	2·9 (5·3)	2·5 (4·0)	2·4 (4·1)
Previous use of antihyperglycaemic medications	23 (17%)	20 (15%)	20 (15%)	19 (14%)
Bodyweight, kg	99·6 (23·3)	95·1 (21·9)	99·6 (25·1)	93·3 (18·6)
BMI, kg/m ²	36·6 (7·4)	35·6 (6·7)	36·4 (7·1)	34·7 (6·5)
Waist circumference, cm	113·7 (15·9)	110·4 (13·9)	113·3 (16·6)	109·2 (13·9)
Triglycerides, mg/dL	176·9 (138·3)	174·2 (124·8)	172·7 (102·9)	174·7 (105·6)
Non-HDL cholesterol, mg/dL	135·3 (39·0)	138·4 (39·0)	139·4 (38·8)	142·9 (40·4)
Blood pressure, mm Hg				
Systolic	124·4 (10·0)	125·7 (11·0)	125·4 (10·9)	124·2 (9·9)
Diastolic	79·6 (7·0)	79·2 (7·8)	79·3 (6·2)	79·0 (6·8)
eGFR, mL/min per 1·73 m ²	85·0 (22·5)	88·3 (22·4)	86·7 (22·1)	87·0 (23·2)

Data are mean (SD) or n (%). Randomly allocated population. HbA_{1c}=glycated haemoglobin. eGFR=estimated glomerular filtration rate.

Table 1: Baseline demographics and clinical characteristics



(41–64% vs 3%; figure 3D). Similar results were observed for the efficacy estimand (appendix pp 25–26, 44). Exploratory subgroup analyses for change in bodyweight are presented in the appendix (pp 45–46).

Changes from baseline in waist circumference and BMI are presented in the appendix (p 47). Triglycerides decreased from baseline to week 40 by 26.7% (SE 3.0) with retatrutide 4 mg, 34.1% (3.0) with 9 mg, and 27.0% (3.2) with 12 mg, versus 3.4% (4.1) with placebo. Non-HDL cholesterol decreased by 15.6% (SE 2.3) with retatrutide 4 mg, 17.0% (2.3) with 9 mg, and 15.5% (2.1) with 12 mg, versus 3.8% (2.7) with placebo (figure 3E). From a mean baseline of 124.9 mm Hg, at week 40, systolic blood pressure decreased by 5.0 mm Hg (SE 0.9) with retatrutide 4 mg, 4.7 mm Hg (0.8) with 9 mg, and 5.4 mm Hg (0.8) with 12 mg, versus 1.5 mm Hg (0.7) with placebo (figure 3F). Changes in additional cardiometabolic parameters are shown in the appendix (pp 23–27).

Treatment-emergent adverse events occurred in 84 (63%) of 134 participants in the retatrutide 4 mg group, 85 (64%) of 133 participants in the retatrutide 9 mg group, 85 (63%) of 136 participants in the retatrutide 12 mg group, and 77 (57%) of 134 participants in the placebo group (table 2). The most common treatment-emergent adverse events reported in retatrutide-treated participants were gastrointestinal, mostly mild to moderate in severity, and occurred primarily during the dose-escalation period (appendix pp 28, 48). Study intervention discontinuations due to adverse events occurred in three (2%) participants in the retatrutide 4 mg group, six (5%) in the 9 mg group, and seven (5%) in the 12 mg group, and in no participants in the placebo group (appendix p 29).

Serious adverse events occurred in a total of 14 (3%) participants: six (4%) in the retatrutide 4 mg group, one (1%) in the 9 mg group, five (4%) in the 12 mg group, and two (1%) in the placebo group (table 2). Two deaths occurred during the trial, both of which were in the retatrutide 4 mg group and were not related to study intervention (appendix p 29). Clinically significant

(level 2) hypoglycaemia (blood glucose <54 mg/dL [<3.0 mmol/L]) was reported by two (2%) participants in the retatrutide 9 mg group and one (1%) participant in the retatrutide 12 mg group. Severe hypoglycaemia was not reported by any participants in the study (table 2). Rescue medication for severe hyperglycaemia was used by three (2%) participants in the retatrutide 4 mg group, nine (7%) in the 9 mg group, and two (1%) in the 12 mg group, compared with 26 (19%) participants in the placebo group.

Dysesthesia events occurred in 15 (4%) participants treated with retatrutide. Sensitive skin was the most frequently reported event (appendix p 30). Generally, the severity and duration of the dysesthesia events were mild to moderate, and the majority resolved while the participants were on study intervention. Three participants reported severe dysesthesia (one in each retatrutide treatment group), resulting in two of these participants discontinuing study intervention (appendix p 30).

At week 40, changes in mean pulse rate were -0.1 bpm (SE 0.6) with retatrutide 4 mg, 1.5 bpm (0.6) with 9 mg, and 0.9 bpm (0.6) with 12 mg, compared with -1.1 bpm (0.6) with placebo (appendix pp 31, 49). Seven participants treated with retatrutide reported arrhythmias or cardiac conduction disorders (table 2). One of these participants in the retatrutide 4 mg group, who had a medical history of coronary artery disease, had a cardiac arrest unrelated to study drug, which led to study intervention discontinuation. The other events did not cause interruption to study intervention.

There were no cases of positively adjudicated pancreatitis in the trial. Changes in pancreatic amylase and lipase, and hepatic alanine aminotransferase and aspartate aminotransferase are shown in the appendix (p 32). Shifts in alanine aminotransferase and aspartate aminotransferase categories are shown in the appendix (pp 33–35). There were no suspected cases of drug-induced liver injury. No cases of medullary thyroid cancer were reported.

At baseline, 25 (5%) participants had diabetic retinopathy diagnosed based on fundus photography assessment (appendix p 36). Among all participants assessed after baseline, nine (2%) participants had worsening of diabetic retinopathy in at least one eye. Of those who had diabetic retinopathy at baseline, one participant in the retatrutide 9 mg group and two participants in the placebo group had worsening at any time after baseline (appendix p 36).

Discussion

TRANSCEND-T2D-1 showed efficacy and safety of the once-weekly GIP, GLP-1, and glucagon receptor agonist retatrutide compared with placebo in people with type 2 diabetes treated with diet and exercise alone. All doses of once-weekly retatrutide (4 mg, 9 mg, and 12 mg) were superior to placebo in reducing HbA_{1c} concentration at week 40, with reductions of up to 1.9% (21.2 mmol/mol).

Figure 2: Effects of retatrutide on glycaemic control

Data are model-based estimate unless otherwise noted. Error bars represent SE. Arrows indicate when the randomly assigned doses of retatrutide (4 mg, 9 mg, and 12 mg) were reached. Only p values controlled for multiplicity (family-wise type I error) are shown. CIs were not adjusted for multiplicity and should not be used for hypothesis testing. (A) Change from baseline in HbA_{1c} at 40 weeks from ANCOVA with multiple imputation by treatment (treatment regimen estimand). (B) Actual HbA_{1c} over 40 weeks from MMRM analysis (efficacy estimand). (C) Change from baseline in FSG at 40 weeks from ANCOVA with multiple imputation (treatment regimen estimand). (D) Actual FSG over 40 weeks from MMRM analysis (efficacy estimand). (E) Proportion of participants reaching the indicated HbA_{1c} targets at 40 weeks from logistic regression with multiple imputation (treatment regimen estimand) with p values for risk difference. (F) 7-Point SMBG profiles and daily mean SMBG at baseline and week 40 from MMRM (efficacy estimand). ETD=estimated treatment difference. FSG=fasting serum glucose. HbA_{1c}=glycated haemoglobin. MMRM=mixed-model repeated measures. SMBG=self-monitored blood glucose.

Figure 3: Effects of retatrutide on bodyweight, lipids, and systolic blood pressure

Data are model-based estimate unless otherwise noted. Error bars represent SE.

Arrows indicate when the randomly assigned doses of retatrutide (4 mg, 9 mg, and 12 mg) were reached. Only p values controlled for multiplicity (family-wise type I error) are shown. CIs were not adjusted for multiplicity and should not be used for hypothesis testing.

(A) Percentage change from baseline in bodyweight at 40 weeks from ANCOVA with multiple imputation (treatment regimen estimand).

(B) Percentage change from baseline in bodyweight over 40 weeks from mixed-model repeated measures (efficacy estimand).

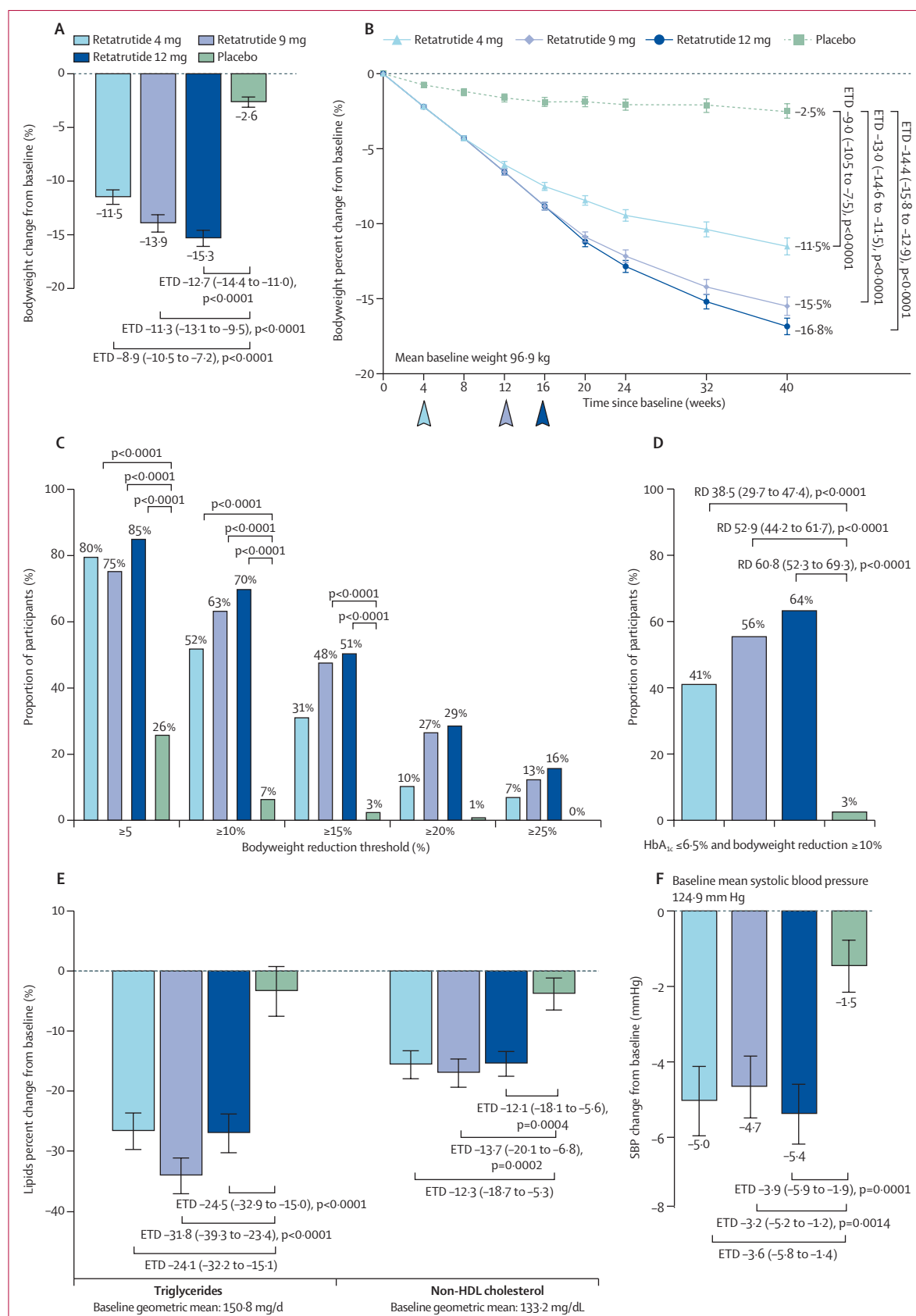
(C) Proportion of participants reaching the indicated bodyweight reduction thresholds at 40 weeks from logistic regression with multiple imputation (treatment regimen estimand) with p values for risk difference.

(D) Proportion of participants reaching the composite target of HbA_{1c} less than or equal to 6.5% (48 mmol/mol) and 10% or more bodyweight reduction from baseline at 40 weeks from logistic regression with multiple imputation (treatment regimen estimand) with risk difference.

(E) Percentage change from baseline in serum triglycerides and non-HDL cholesterol at 40 weeks from ANCOVA with multiple imputation (treatment regimen estimand).

(F) Change from baseline in systolic blood pressure at 40 weeks from ANCOVA with multiple imputation (treatment regimen estimand).

ETD=estimated treatment difference. HbA_{1c}=glycated haemoglobin. RD=risk difference.



The HbA_{1c} target of less than 7·0% was reached by 82–89% of participants treated with retatrutide, and the target of less than or equal to 6·5% was reached by 75–83%. Furthermore, HbA_{1c} levels of less than 5·7%, representing a normal value for HbA_{1c}, were reached by 35–40% of participants treated with retatrutide, without any case of severe hypoglycaemia. Reductions in HbA_{1c} did not appear to reach a plateau by the end of the 40-week treatment period in the retatrutide 9 mg and 12 mg groups.

Improvements in glycaemia with retatrutide were driven by improvements in both fasting and postprandial hyperglycaemia based on the central laboratory fasting serum glucose data and 7-point SMBG profiles. This finding is consistent with the results of the phase 2 study of retatrutide and confirms the optimised pharmacology of GIP, GLP-1, and glucagon receptor agonism conferred by retatrutide on glycaemic control.^{20,21} Although dual GIP and GLP-1 receptor agonism has been shown to improve glycaemic control by enhancing glucose-dependent insulin secretion and suppressing glucagon secretion, delaying gastric emptying, and improving insulin sensitivity,^{10,25,26} additional research on the role of glucagon agonism in the setting of hyperglycaemia or normoglycaemia is warranted to further understand the role of glucagon agonism in glycaemic control.

Clinically meaningful bodyweight reduction was observed with retatrutide in TRANSCEND-T2D-1. The American Diabetes Association standards of care suggest a minimum of 5% weight reduction for meaningful improvements in glycaemia and other cardiometabolic parameters like lipids and blood pressure.⁷ Clinical improvements are progressive, with more than 10% weight loss associated with reduced mortality and need for cardiometabolic pharmacotherapy, and more than 15% weight loss maximises the metabolic benefit with potential for modifying the pathophysiology of type 2 diabetes.^{7,8} At the 12 mg dose, retatrutide showed an average of 15·3% weight reduction with no evidence of reaching a plateau, suggesting that a substantial proportion of individuals with early type 2 diabetes could reach this threshold with long-term treatment. Reduction in bodyweight was also associated with reduction in waist circumference, which suggests a reduction in visceral adipose fat. Additional clinical studies are warranted to assess whether weight reduction with retatrutide results in clearance of ectopic fat, especially in the liver and pancreas, to break the twin cycles that are hypothesised to fuel type 2 diabetes progression.²⁷

Given the importance of early glycaemic control and concurrent weight reduction in altering the trajectory of type 2 diabetes,^{8,28,29} a composite endpoint of HbA_{1c} of 6·5% or less and weight reduction of 10% or more was assessed in TRANSCEND-T2D-1. Composite endpoints of this kind more comprehensively capture overall treatment benefit than either measure alone, and their adoption in diabetes trials should be considered. Up to 64% of

participants treated with retatrutide reached the composite endpoint at week 40, showing the efficacy of retatrutide in improving glycaemic control while simultaneously driving clinically meaningful weight reduction.

Although cross-trial comparisons are restricted by differences in study design, in SUSTAIN-1—a

	Retatrutide 4 mg (n=134)	Retatrutide 9 mg (n=133)	Retatrutide 12 mg (n=136)	Placebo (n=134)
Participants with one or more treatment-emergent adverse events	84 (63%)	85 (64%)	85 (63%)	77 (57%)
Serious adverse events	6 (4%)	1 (1%)	5 (4%)	2 (1%)
Deaths*	2 (1%)	0	0	0
Adverse events leading to study intervention discontinuation	3 (2%)	6 (5%)	7 (5%)	0
Gastrointestinal adverse events leading to study intervention discontinuation	0	1 (1%)	4 (3%)	0
Upper abdominal pain	0	0	1 (1%)	0
Constipation	0	0	1 (1%)	0
Dyspepsia	0	0	1 (1%)	0
Nausea	0	1 (1%)	1 (1%)	0
Treatment-emergent adverse events occurring in 5% or more of participants in any treatment group				
Diarrhoea	25 (19%)	35 (26%)	31 (23%)	6 (4%)
Nausea	22 (16%)	26 (20%)	36 (26%)	5 (4%)
Vomiting	21 (16%)	20 (15%)	24 (18%)	3 (2%)
Hyperglycaemia	5 (4%)	10 (8%)	5 (4%)	35 (26%)
Decreased appetite	10 (7%)	15 (11%)	11 (8%)	2 (1%)
Headache	5 (4%)	8 (6%)	7 (5%)	13 (10%)
Abdominal distension	7 (5%)	8 (6%)	13 (10%)	3 (2%)
Constipation	6 (4%)	11 (8%)	10 (7%)	1 (1%)
Asthenia	5 (4%)	5 (4%)	5 (4%)	9 (7%)
Fatigue	1 (1%)	8 (6%)	10 (7%)	4 (3%)
Pain	3 (2%)	5 (4%)	4 (3%)	11 (8%)
Abdominal discomfort	6 (4%)	10 (8%)	4 (3%)	1 (1%)
Nasopharyngitis	6 (4%)	2 (2%)	4 (3%)	9 (7%)
Upper abdominal pain	7 (5%)	1 (1%)	5 (4%)	7 (5%)
Back pain	4 (3%)	3 (2%)	2 (1%)	10 (7%)
Eruetation	1 (1%)	2 (2%)	7 (5%)	0
Other treatment-emergent adverse events of interest				
Hypoglycaemia (blood glucose <54 mg/dL)	0	2 (2%)	1 (1%)	0
Severe hypoglycaemia	0	0	0	0
Initiation of rescue therapy for severe hyperglycaemia	3 (2%)	9 (7%)	2 (1%)	26 (19%)
Arrhythmias and cardiac conduction disorders	4 (3%)	2 (2%)	1 (1%)	0
Malignancies	3 (2%)	0	0	1 (1%)
Adjudication-confirmed major adverse cardiovascular events	3 (2%)	0	0	0
Cholelithiasis	0	1 (1%)	3 (2%)	0
Dysesthesia	6 (4%)	3 (2%)	6 (4%)	0
Injection site reactions	2 (1%)	2 (2%)	2 (1%)	1 (1%)

Data are n (%). Safety analysis set. Participants might be counted in more than one category. Number of episodes was reported if available. MedDRA version 28.1. *Deaths are also included as serious adverse events and discontinuations due to adverse events.

Table 2: Adverse events

monotherapy trial of a selective GLP-1 receptor agonist in adults with type 2 diabetes—semaglutide 1.0 mg lowered mean HbA_{1c} to approximately 6.6% at 30 weeks (mean baseline of 8.1%), with bodyweight reduction of approximately 5% (efficacy estimand).¹² In SURPASS-1, a monotherapy study for a dual GIP and GLP-1 receptor agonist, treatment with tirzepatide 15 mg lowered mean HbA_{1c} to approximately 5.9% at 40 weeks (mean baseline of 7.9%), with 11% bodyweight reduction (efficacy estimand).¹³ In the present study, retatrutide 12 mg, a triple GIP, GLP-1, and glucagon receptor agonist, lowered mean HbA_{1c} to approximately 6.0% at 40 weeks (mean baseline 7.9%) with 17% bodyweight reduction (efficacy estimand), with no evidence of plateau in the weight loss trajectory. These data suggest that selective GLP-1 receptor agonism could reach an apparent glycaemic floor around an HbA_{1c} of 6.5%, whereas agents incorporating GIP co-agonism drive HbA_{1c} towards near-normoglycaemic levels below 6%. Retatrutide appears to preserve this degree of glycaemic lowering while producing meaningfully greater weight reduction, potentially attributable to the additional glucagon receptor activation, which might increase energy expenditure and fatty acid oxidation and impart an additive effect on lowering food intake. These indirect observations are hypothesis generating; differences in diabetes duration, baseline bodyweight, and study design preclude definitive conclusions, but head-to-head trials are ongoing and will establish relative efficacy and safety of retatrutide versus semaglutide and tirzepatide (NCT06260722 and NCT06662383).

Retatrutide treatment also improved blood pressure and lipid profiles. Although GIP and GLP-1 receptor agonism has been shown to improve lipid parameters, this effect might be further potentiated by involvement of glucagon agonism, which has established effects in stimulating hepatic β oxidation of fatty acids and inhibiting hepatic lipogenesis.³⁰

The safety profile of retatrutide was overall consistent with other molecules with GLP-1 agonist activity. The most common treatment-emergent adverse events were mostly mild-to-moderate gastrointestinal events that subsided over time. Study intervention discontinuation due to gastrointestinal events was low (1%) across all participants treated with retatrutide. The optimised retatrutide dose-escalation scheme being implemented in the phase 3 programme, starting at a 2 mg dose and gradually increasing the dose up to 12 mg, appeared to be well tolerated in adults with type 2 diabetes and resulted in less study intervention discontinuation due to gastrointestinal events compared with the phase 2 study (3%).²¹ The incidence of serious adverse events was low across all retatrutide dose groups and did not suggest an emergent safety signal. No severe hypoglycaemia events occurred during the trial. Incidence of dysesthesia was 2–4% for participants treated with retatrutide, and the majority of these events resolved while the participants were on study drug.

Strengths of this trial included the double-blind, randomised, placebo-controlled trial design, which restricts bias in adverse event reporting and adherence behaviours. The high study completion rates and low discontinuation rates resulted in minimal missing data and support the tolerability of retatrutide treatment for people with type 2 diabetes. The monotherapy trial design reduced the chance of bias being introduced by changes in background antihyperglycaemic agents. Additionally, the short mean duration of diabetes of 2.5 years allowed the evaluation of retatrutide early after the diagnosis of type 2 diabetes.

A limitation of the trial is the largely medication-naïve participant population, which reduces the generalisability of the study findings to the broader population of people with type 2 diabetes who might be on prescription antihyperglycaemic medications or insulin regimen. Participants from only three countries were included in the trial, which also reduces generalisability to broader populations. Additionally, the study treatment duration of 40 weeks presents a limitation, as a longer treatment duration could allow longer-term safety of retatrutide to be assessed and the full extent of efficacy (such as HbA_{1c} and weight reduction plateaus) to be reached. These limitations might be addressed by the additional ongoing TRANSCEND-T2D trials: TRANSCEND-T2D-2, investigating retatrutide compared with semaglutide over 80 weeks in participants using metformin with or without an SGLT2 inhibitor; and TRANSCEND-T2D-3, investigating retatrutide compared with placebo over 52 weeks in participants with type 2 diabetes and renal impairment.

In conclusion, the results of the TRANSCEND-T2D-1 clinical trial suggest that once-weekly retatrutide could offer first-line treatment for early type 2 diabetes without background antihyperglycaemic medication, providing effective glycaemic control in addition to benefits of clinically relevant weight reduction and improvement in cardiometabolic parameters with no severe hypoglycaemia events reported in the study, as well as gastrointestinal side-effects and an adverse event profile consistent with other therapies with GLP-1 agonist activity.

Contributors

The sponsor (Eli Lilly and Company) conceived, designed, and supervised the study, which included site monitoring, data collection, and analysis. HP, AB, F-ZJ, and BL contributed to the study design. HP, AB, and YC provided medical oversight during the trial. F-ZJ, BL, and RL were responsible for the data curation, formal analysis, validation, visualisation, and editing of statistical analyses. HP, AB, YC, F-ZJ, BL, and RL contributed to drafting the manuscript. HSB, MW, F-ZJ, YC, HP, and AB accessed and verified the data underlying the study. All authors participated in interpretation of the data and critical review of the manuscript, approved the final and previous versions of the manuscript, had full access to all of the data in the study, and had final responsibility for the decision to submit for publication.

Declaration of interests

HSB reports research support (paid to institution) from Abbott, Amgen, AstraZeneca, Boehringer Ingelheim, Eli Lilly, Ionis, Novartis, Novo

Nordisk, Pfizer, Roche, and Vertex; and serving as a speaker and on advisory panels for Eli Lilly, Novo Nordisk, and Roche. MW reports payment as a member of a speakers bureau from, support for attending meetings or travel from, and stocks in Eli Lilly and Company. EL reports consulting and speaker fees from Abbott, Eli Lilly, Dexcom, CeQur, Insulet, Madrigal Pharmaceuticals, and Mannkind. PS reports speaker fees from Eli Lilly and Company. F-ZJ, BL, RL, YC, HP, and AB are employees of and shareholders in Eli Lilly and Company.

Data sharing

Eli Lilly and Company provides access to all individual participant data collected during the trial, after anonymisation, with the exception of pharmacokinetic or genetic data. Data are available to request 6 months after the indication studied has been approved in the USA and EU and after primary publication acceptance, whichever is later. No expiration date of data requests is currently set once data are made available. Access is provided after a proposal has been approved by an independent review committee identified for this purpose and after receipt of a signed data sharing agreement. Data and documents, including the study protocol, statistical analysis plan, clinical study report, and blank or annotated case report forms, will be provided in a secure data sharing environment. For more on submitting a request see <http://www.vivli.org>.

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